Day 46 – SFTP Sender Adapter

Class date: 3rd September 2026 | Module: Adapters (SFTP Sender), Non-Auth, Read Lock Strategy

Two corrections/clarifications to your handwritten notes:
1) You wrote **"why HTTP Adapter doesn't have Proxy Type."** To be precise: it's specifically the **HTTP Sender Adapter** that has no Proxy Type – the **HTTP Receiver Adapter absolutely does** have Proxy Type (Internet/On-Premise), as covered back on Day 42. Your own reasoning underneath (push vs. pull/push) is exactly right, it just needed the word "Sender" attached to be fully correct.
2) For the **Rename** Read Lock Strategy, you noted "it will rename the file and pick that file, the original file will be there itself." Actually the file is renamed *in place* on the SFTP server – there is no separate original left behind under the old name; CPI reads the file under its new, renamed name. Corrected in Section 4 below.

1. Recap: Non-Authentication (Non-Auth) usage (start here first)

Following up on a previous **POC (Proof of Concept)** discussion: the key question is, if you want to call an **on-premise system**, how do you actually call it?

- **The primary/only way** to reach an on-premise system is via the **Cloud Connector** (Days 44–45).
- **An alternative pattern** is calling the on-premise system using **Non-Auth ("None")** as the adapter's own authentication setting, while handling authentication *manually* inside the iFlow: you first make a call inside the iFlow to fetch a **Bearer/OAuth token**, and then hit the on-premise endpoint yourself by explicitly setting the **Authorization header** with that token (rather than letting the adapter's built-in Authentication dropdown manage it).

Still required either way: when going through the Cloud Connector, you need the **Virtual Host name** and **Location ID** (if multiple Cloud Connectors are registered). The **BTP Admin team** needs the **Virtual Host, Virtual Port, and Location ID** in order to connect to the on-premise SAP system, along with **Internal Host, Internal Port, Protocol,** and the **type of system** being exposed (e.g. Non-SAP System) – this is the same System Mapping information covered on Day 45.

2. Why the HTTP Sender Adapter has no Proxy Type

Proxy Type (Internet / On-Premise) only makes sense for an adapter that **initiates** an outbound call – i.e. one that has to decide *where* to reach out to. The difference comes down to the **triggering point**:

- **HTTP (Sender side):** CPI simply **exposes a URL** and waits to be called. It only **pushes** data *to* whoever calls that URL – it never has to reach out anywhere itself, so there is nothing for Proxy Type to configure.
- **SFTP (both Sender and Receiver):** SFTP always involves CPI actively connecting *out* to a server – either to **pull** files from a source SFTP server, or to **push** files to a target SFTP server. Because CPI is always the one initiating the connection, it always needs to specify Proxy Type (Internet/On-Premise) so it knows how to reach that server.

That's why **HTTP Sender** has no Proxy Type field, while **SFTP Sender, SFTP Receiver, and the HTTP Receiver** all do.

3. Scheduler (brief recap)

Scheduler options for the SFTP Sender adapter: **Schedule on Day** (run at a specific calendar date/time), **Schedule to Recurring** (repeat on an interval, e.g. every N minutes/hours), and **Advanced** (custom cron-style expressions for more complex scheduling needs).

4. Processing tab – Empty File Handling & Read Lock Strategy

Empty File Handling

- **Process Empty File** – the empty (0 KB) file is picked up and processed through the iFlow like any other file.
- **Skip Empty File** – the empty file is simply ignored; no message is created at all.
- **MPL with Post-Processing Only** – no message is sent into the main integration logic, but an entry is still created in the **Message Processing Log (MPL)** and post-processing (e.g. delete/move) still runs on the file – useful for keeping an audit trail of empty files without triggering real processing.

Real-time relevance, as flagged in class: an empty file (0 KB) reaching the mapping step will usually cause a **mapping error**, and the whole flow execution gets disturbed/fails. This is exactly why Empty File Handling needs to be configured deliberately rather than left to fail at runtime.

Read Lock Strategy – why it's needed

Scenario: the source system is still writing the file's content when CPI polls the SFTP server. If CPI picks up the file mid-write, it processes an incomplete file. **Read Lock Strategy** is the solution – it makes CPI wait until the source system has fully finished writing before the file is picked up.

The 3 real Read Lock Strategy options (besides the default, None):

- **Content Changed** – CPI repeatedly scans the file to see if its content is still changing. **Change Lock Check Interval (in ms)** controls how often it re-checks – e.g. **5,000 ms (5 sec)**. Only once the content stops changing between two consecutive scans is the file considered final and picked up.
- **Done File Expected** – CPI waits for a companion "done" file as the signal that writing is complete. Example: for **file.txt**, the companion file would be **file.done**; only once *both* file.txt and file.done exist in the folder does CPI pick up file.txt. The done-file name pattern is typically set as **${file:name}.done**.
- **Rename** – CPI renames the file on the SFTP server *before* reading it, then reads it under its new name. *(Corrected above: this changes the file's name in place – the file does not remain accessible under its original name once renamed.)*

> The **Read Lock Check Interval's timeout** is **4 times the interval value** you configure – this is the maximum time CPI will wait to acquire the lock before giving up on that file for this poll cycle.

5. Worker nodes – parallel vs. sequential polling

A CPI tenant can have **multiple worker nodes** running the same deployed iFlow, each independently polling the SFTP server. Example given in class: if **4 files** are waiting and **4 worker nodes** are available, each node can pick up and process one file **in parallel**.

- **Poll on One Node Only** (checkbox) – when **checked**, only a single worker node performs the polling at a time, so files are processed in **serial/sequential order**. This matters when file processing order is important, or when you want to limit how many parallel connections hit the SFTP server.
- When **unchecked**, multiple worker nodes can poll and process files **in parallel**, independently of each other.

Stop on Exception: in **Sequential** mode (Poll on One Node Only checked), if one file fails, processing of further files **stops**. In parallel mode (unchecked), files continue to be processed independently even if one fails.

Sorting – when processing needs a defined order, files can be sorted (ascending/descending) by: **File Name, File Size,** or **File Timestamp.**

6. Other Processing tab fields

Field	Notes
Max. Messages per Poll	How many messages/files are gathered per poll cycle. Range 1–500, default 20. (If multiple worker nodes are active, the effective total is this value × number of nodes.)
Lock Timeout (in min)	How long CPI waits before attempting to reprocess a file after an outage, default noted as 15 min – should be set higher than the time it takes to process a full poll's worth of files, to avoid double-processing.
Use Fast Check	Performs the file-exists check directly on the server for speed; enabled by default.
Include Subdirectories	Looks for matching files inside all nested subfolders too, e.g. scans /output, /output/sample, /output/sample/test, /output/sample/test/temp, etc., not just the top-level folder.
Change Directories Stepwise	Not used in this demo (unchecked/ignored) – changes directory levels one at a time rather than jumping straight to the deepest path.

Quick-recall summary

- Reach an on-premise system either via **Cloud Connector** (standard way), or via **Non-Auth + manual Bearer token in the iFlow** (alternative pattern).
- **HTTP Sender** has no Proxy Type (it only exposes/pushes); **HTTP Receiver, SFTP Sender, SFTP Receiver** all have Proxy Type (they actively reach out).
- **Empty File Handling:** Process Empty File / Skip Empty File / MPL with Post-Processing Only. Empty files reaching mapping usually cause errors.
- **Read Lock Strategy:** Content Changed, Done File Expected, Rename – all prevent picking up a file that's still being written. Lock timeout = 4 × check interval.
- **Poll on One Node Only** = sequential/serial processing across one worker; unchecked = parallel across multiple worker nodes.
- **Max Messages per Poll** default 20 (1–500); **Sorting** by File Name/Size/Timestamp.